

Project Report

Laser Tripwire Security Alarm System

Subject: Arduino-Based Security Systems

Project Category: Embedded Systems & Robotics

Project Image Placeholder

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Arduino-Based Laser Tripwire Security Alarm System

1. Introduction

In today's world, security is a paramount concern for residential, commercial, and industrial facilities. Traditional security systems can be expensive, complex to install, and prone to high power consumption. Microcontroller-based embedded systems present a cost-effective, energy-efficient, and highly customizable alternative.

This project details the design, construction, and programming of an Arduino-Based Laser Tripwire Security Alarm System. Using the properties of light-dependent resistors (LDRs) and low-power laser diodes, this system creates an invisible boundary line (tripwire). If a human or object crosses this line, an alarm is instantly triggered. This project offers a hands-on introduction to sensor integration, analog-to-digital signals, and real-time event-driven programming.

2. Problem Statement

Securing restricted perimeter areas (such as doorways, corridors, or boundary walls) requires continuous monitoring. Typical passive infrared (PIR) sensors detect motion but are prone to false positives caused by pets, thermal fluctuations, or wind. On the other hand, commercial active infrared barriers are highly expensive.

There is a clear need for a low-cost, highly precise, easily deployable, and localizable intrusion detection system that:

- Operates under minimal power.
 - Pinpoints the exact entry crossing point.
 - Can be constructed in a short period (under 24 hours) using modular components.
 - Provides immediate audio-visual alerts upon intrusion.
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3. Proposed Solution

The proposed solution implements a laser-focused optical barrier. By aligning a low-cost laser diode to point directly at a Light Dependent Resistor (LDR) housed inside a light-shielding tube, we construct a reliable optical switch.

- Normal State: The laser continuously shines on the LDR, keeping its electrical resistance extremely low.
 - Intrusion State: An intruder cuts the beam. The light intensity on the LDR drops instantly, causing its resistance to skyrocket.
 - Action: The Arduino detects this sudden change in resistance via an analog voltage divider circuit, immediately sounding a piezoelectric buzzer and flashing warning LEDs.
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4. Project Information

- Microcontroller Platform: Arduino Uno (Atmega328P)
 - Estimated Build Time: 1 to 2 Hours
 - Programming Language: C++ (Arduino IDE)
 - Power Source: 9V DC Battery / USB Power bank
 - Total Estimated Cost: ~10 - 15 USD
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5. Design Details

5.1 Component List & Specifications

Component	Qty	Function / Specification
Arduino Uno R3	1	Brain of the system. Reads inputs, runs logic, controls outputs.
Laser Diode (650nm, Red)	1	Emits a focused light beam. Power: 5mW, Voltage: 5V.
LDR (Photoresistor)	1	Light sensor. Resistance decreases as light intensity increases.
10kΩ Resistor	1	Pull-down resistor forming a voltage divider with the LDR.
220Ω Resistor	2	Current-limiting resistors for the warning LEDs.
Buzzer (5V Active/Passive)	1	Audio indicator. Emits alarm tone upon breach.
Red LED	1	Visual alarm indicator (flashes when tripped).
Green LED	1	Visual status indicator (constant glow when system is safe).
Breadboard & Jumper Wires	1	Used for solderless prototyping and connections.

Component Visuals

These visuals show every major part needed to build the laser tripwire security alarm.



Arduino Uno R3

Microcontroller board



Laser Diode

650nm red beam source



LDR Sensor

Detects beam brightness



10kΩ Resistor

Voltage divider pull-down



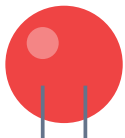
220Ω Resistors

LED current limiters



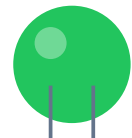
Buzzer

Audio alarm output



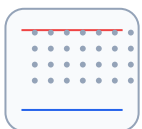
Red LED

Alarm/tripped indicator



Green LED

Safe/armed indicator



Breadboard

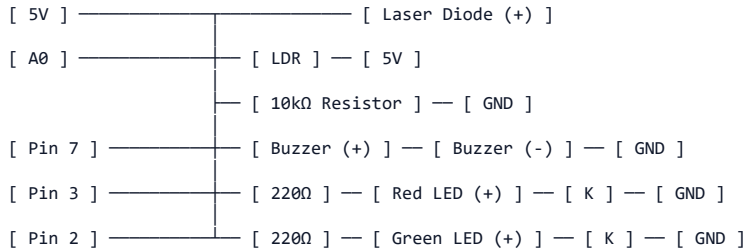
Solderless prototype base



Jumper Wires + Power

Connections and 9V/USB supply

5.2 Circuit Connections (Wiring Diagram)



Circuit / Assembly Image Placeholder

Add wiring photo, schematic, or breadboard image here

- Note: The Laser Diode can be powered directly from the Arduino's 5V pin, or hooked up to an external battery if placed across a wide room.
- Light Shielding Tip: Cover the sides of the LDR with a small piece of dark straw or electrical tape (forming a small tube). This blocks ambient room light from reaching the LDR, ensuring only the laser beam hits it directly.

6. Working / Methodology

The logical flow of the Laser Tripwire System operates in a continuous loop:

Start system
Measure ambient light and calibrate
Read analog value from LDR A0
Decision: is the value below threshold?
No: keep green LED on, alarm off, and continue monitoring
Yes: turn green LED off, flash red LED, trigger buzzer, then await reset

Mathematical Working

The analog pin A0 reads voltage using a Voltage Divider equation:

$$V_{out} = V_{in} \times \left(\frac{R_{pull-down}}{R_{LDR} + R_{pull-down}} \right)$$

1. When Laser is ON the LDR: R_{LDR} is very low (approximately 100 ohm). Therefore, V_{out} is close to 5V. The Arduino reads a high analog value (above 800 out of 1023).
2. When Laser is BLOCKED: R_{LDR} spikes dramatically (approximately 10kohm - 1Mohm). V_{out} drops significantly. The Arduino reads a low analog value (typically below 400).

7. Project Description & Implementation

7.1 Complete Arduino Source Code

You can copy this code directly into your Arduino IDE, compile, and upload:

```
/**
 * Arduino Laser Tripwire Alarm System
 * -----
 * This code reads an LDR voltage output. If the light intensity drops below
 * the calibrated threshold, it triggers an audible buzzer and red LED.
 */

// Pin Definitions
const int ldrPin = A0;      // Analog input pin connected to the LDR
const int greenLed = 2;     // Digital output pin for Safe state (Green LED)
const int redLed = 3;       // Digital output pin for Alarm state (Red LED)
const int buzzerPin = 7;    // Digital output pin for Active Buzzer

// System Calibration Variables
int threshold = 600;        // Voltage threshold (adjustable based on room brightness)
bool isTripped = false;    // Variable to track alarm status

void setup() {
  // Initialize Serial Monitor for calibration support
  Serial.begin(9600);

  // Configure I/O Pins
  pinMode(greenLed, OUTPUT);
  pinMode(redLed, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  // Initialize Pin States
  digitalWrite(greenLed, HIGH); // Start with green LED on (System Armed)
  digitalWrite(redLed, LOW);
  digitalWrite(buzzerPin, LOW);

  Serial.println("System Calibrating... Alignment complete.");
}

void loop() {
  // Read current analog value from LDR
  int ldrValue = analogRead(ldrPin);

  // Output to Serial Monitor for testing and threshold calibration
  Serial.print("Current LDR Light Reading: ");
  Serial.println(ldrValue);

  // Logic decision based on light levels
  if (ldrValue < threshold) {
    // Laser beam has been blocked!
    triggerAlarm();
  } else {
    // Laser beam is intact
    resetSystem();
  }

  delay(50); // Small delay to stabilize readings
}

// Function to activate alert states
void triggerAlarm() {
  digitalWrite(greenLed, LOW); // Turn off safe indicator

  // Flash red LED and pulse buzzer to create a siren effect
  digitalWrite(redLed, HIGH);
  tone(buzzerPin, 1000); // Produce a 1KHz tone
  delay(150);

  digitalWrite(redLed, LOW);
  noTone(buzzerPin);
  delay(100);
}

// Function to return system to armed state
void resetSystem() {
  digitalWrite(greenLed, HIGH);
  digitalWrite(redLed, LOW);
  noTone(buzzerPin);
}
```

Project Demonstration Image Placeholder

Add final working model or test photo here

8. Applications of the Project

- Home Perimeter Security: Can be placed across entry gates, corridors, or window sills to detect unauthorized access.
- Industrial Safety Barriers: Placed around high-risk automated machinery. If a worker crosses the laser grid, the system automatically shuts off the machine to prevent injury.
- Museum Exhibit Protection: Creating invisible boxes around valuable artifacts. Tripping the laser triggers a silent alarm at the security desk.
- Visitor/Object Counter: By incrementing a variable in code whenever the beam is cut, this circuit can count visitors entering a room or products moving along a conveyor belt.

9. Conclusion

The Laser Tripwire Security System successfully demonstrates the integration of optical sensing with real-time microcontroller logic. The system acts instantly upon detecting a beam breach, providing robust and reliable localized perimeter defense. Its modular nature allows for easy expansions, such as integrating Wi-Fi modules (ESP8266) to send push notifications to a smartphone when tripped.

10. Acknowledgements

We would like to express our gratitude to our project coordinator and lab assistants for providing the necessary workspace and debugging tools. We also thank the Arduino open-source community for their excellent libraries and hardware documentation which made rapid prototyping possible.

11. References

1. Arduino official documentation: <https://www.arduino.cc> (<https://www.arduino.cc>)
2. LDR Sensor Datasheet: *Standard GL55 Series Photoresistor Specifications*.
3. Embedded C programming paradigms: *Getting Started with Arduino, 3rd Edition (Massimo Banzi)*.

12. Project Demonstration Video Link

Scan this area after the Supabase-hosted image or video URL is provided. The QR code area is intentionally blank for now.



Supabase image/video URL: _____